

UG54 — Big Picture Review

Chapters 4 – 13 | Data-Driven Investing with Python

Alan Moreira • NYU Stern

Three Pillars of the Course

MEASURE

Chapters 4–5

Returns, risk premia,
Sharpe ratios, timing

DECOMPOSE

Chapters 6–9

Factor models, alpha/beta,
portfolios, estimation

CONSTRUCT

Chapters 10–13

Strategies, allocation,
evaluation, ML

Everything connects through one equation: $w^ = (1/\gamma) \times E[r^e] / \text{Var}(r)$*

The Big Picture

THE IDEAL: If you knew $E[R]$ and Σ , the answer is one equation:
 $W^* = (1/\gamma) \Sigma^{-1} E[R^e] \rightarrow \text{Tangency portfolio} \rightarrow \text{Max Sharpe} \rightarrow \text{Done.}$

REALITY: You don't know $E[R]$ or Σ . Estimation uncertainty changes everything.

Estimate $E[R^e]$ — the hard part

Ch 5.1

Timing:
forecast $E[R]$
with signals

Ch 6

Factor $E[R]$:
 $E[R_i] = \Sigma_k \beta_{ik} \lambda_k$
K premia not N means

Ch 10

Characteristics:
sort $\rightarrow$ long/short
cross-section signals

Estimate Σ — easier but still hard

Ch 5.2

Vol timing:
RV predictable
scale by $1/RV_t$

Ch 6

Factor risk:
 $\Sigma = B \Sigma_{FB}' + \tilde{\Sigma}_{\epsilon}$
fewer parameters

Managing Uncertainty: Ch 7 Diversification reduces risk from Σ errors • Ch 9 Shrinkage & factor structure tame estimation • Ch 11 Heuristics when MV fails • Ch 12 Detect overfitting • Ch 13 Regularization (Lasso, etc.)

Every Chapter Is a Response to Estimation Uncertainty

Ch 4	Returns	Define the object: excess returns, Sharpe ratio — the scoreboard before estimation begins
Ch 5	Timing	Forecast $E[R]$ with signals (hard: OOS $R^2 < 0$) or scale by $1/RV$ (easier: vol is persistent)
Ch 6	Factor Models	Reduce $E[R]$ to K factor premia (not N means). Reduce Σ to factor structure (not N^2 covariances)
Ch 7	Portfolios	Diversification protects you: errors in individual covariances wash out in $W'\Sigma W$
Ch 8	Capital Alloc I	The ideal: $MVE = \Sigma^{-1}E[R^e]$. Two-fund separation holds if you trust your estimates
Ch 9	Estimation	The problem quantified: $\pm 2\sigma$ perturbation in $E[R]$ flips MVE weights. Shrink toward simpler models
Ch 10	Strategies	Side-step: rank stocks on characteristics instead of estimating $E[R]$ directly. Long top, short bottom
Ch 11	Capital Alloc II	When MV fails: $1/N$, risk parity, shrinkage. Simpler heuristics often win out-of-sample
Ch 12	Evaluation	Detect overfitting: SR degrades $>50\%$ OOS. Multiple testing, hold-out discipline, publication bias
Ch 13	ML	Regularize: Lasso beats complex models in low-SNR. Bias-variance tradeoff favors simplicity

Chapter 4

Introduction to Asset Returns

*Big idea: We care about excess returns — compensation for bearing risk.
The Sharpe Ratio is a great first pass. Robust to leverage.*

Returns: The Building Blocks

Key Definitions

Total Return

$$R_t = (P_t + D_t) / P_{t-1} - 1$$

Excess Return

$$R_t^e = R_t - R_t^f$$

Sharpe Ratio

$$SR = E[R^e] / \sigma(R^e)$$

Annualization

$$\mu_A = 12 \times \mu_M \quad \sigma_A = \sqrt{12} \times \sigma_M$$

Why Excess Returns?

Returns (not prices) are scale-free and comparable across assets.

Excess returns strip out the time value of money — what remains is **compensation for risk**.

Example (UNH, daily data):

Ann. mean: 22.4% Vol: 43.3%

Sharpe Ratio ≈ 0.45

Key takeaway: Sharpe ≈ 0.5 is a good benchmark for an annual Sharpe ratio.

Chapter 5

Timing Strategies

*Big idea: Optimal weight depends on expected return AND variance.
You can time on either — but expected returns are much harder to predict.*

The Timing Framework

$$w^* = E[r^e] / (\gamma \times \text{Var}(r))$$

The optimal weight is the risk–return tradeoff scaled by risk aversion

Expected Return Timing (Ch 5.1)

Signal: dividend yield $\rightarrow$ forecast r_{t+h}

$$r_{t \rightarrow t+h} = a + b \times DP_t + u_{t+h}$$

Pitfalls:

- Overlapping returns $\rightarrow$ use HAC (Newey-West)
- In-sample SR = 0.75, but OOS $R^2 < 0$!
- Structural breaks (buybacks) weaken signal

Volatility Timing (Ch 5.2)

$$w_t = c / RV_t \text{ (scale down when vol is high)}$$

Why it works:

- Volatility is highly predictable (persistent)
- Price of risk falls when vol is high
- SR improvement: 0.45 $\rightarrow$ 0.52 (+15%)
- Must lag signal & cap leverage!

Chapter 6

Factor Models

Big idea: Factor models serve two distinct purposes — predicting expected returns (fewer parameters) and modeling risk (structured Σ).

Factor Models as Expected Return Models

From Regression to Expected Returns

Start: $r^e_i = \alpha_i + \beta_{i1} f_1 + \beta_{i2} f_2 + \dots + \beta_{ik} f_k + \epsilon_i$

Take $E[\cdot]$: $E[r^e_i] = \alpha_i + \beta_{i1} \lambda_1 + \beta_{i2} \lambda_2 + \dots + \beta_{ik} \lambda_k$

where $\lambda_k = E[f_k]$ is the risk premium for factor k

The Dimension Reduction

Without factors: estimate N expected returns
(5,000 stocks = 5,000 noisy means!)

With factors: estimate K risk premia + $N \times K$ betas
(6 factors + betas from regression — much easier)

Betas estimated from time-series regressions.
Risk premia λ from factor portfolio returns.

Alpha = What's Left Over

If the model is correct: $\alpha_i = 0$
Expected returns fully explained by factor betas.

If $\alpha_i \neq 0$: mispricing or missing factor.
"Bonuses only for alpha" — the industry mantra.

Example: MSFT daily $\alpha = 0.06\%$ (annualized $\sim 15\%$)
 $\beta_{\text{mkt}} = 1.19 \rightarrow E[r^e] = 1.19 \times 8\% \approx 9.5\%$ from β

Factor Models as Risk Models

From Betas to a Structured Covariance Matrix

$$\Sigma = B \Sigma_F B' + \Sigma_\epsilon$$

$B = N \times K$ beta matrix $\Sigma_F = K \times K$ factor covariance $\Sigma_\epsilon = N \times N$ diagonal (residual variances)

Portfolio risk: $\text{Var}(R_p) = (w'B) \Sigma_F (B'w) + w' \Sigma_\epsilon w$ where $\beta_p = B'w$ is the portfolio's factor exposure

The Parameter Savings

Full Σ : $N(N+1)/2$ parameters

500 stocks $\rightarrow$ 125,250 covariances!

Factor Σ : NK betas + $K(K+1)/2 + N$

500 stocks, 6 factors $\rightarrow$ 3,521 parameters

35× fewer parameters $\rightarrow$ far more stable covariance estimates $\rightarrow$ better portfolios

Strategy Risk Decomposition

Your portfolio w has factor exposure $\beta_p = B'w$

Systematic risk: $\beta_p' \Sigma_F \beta_p$

$\rightarrow$ risk from factor movements (not diversifiable)

Idiosyncratic risk: $w' \Sigma_\epsilon w$

$\rightarrow$ diversifiable with many stocks

$R^2 = \text{systematic} / \text{total variance}$

MSFT vs SPY: $R^2 = 0.42 \rightarrow 42\%$ factor risk

Hedging, Portable Alpha & Appraisal Ratio

Hedged Portfolio

Hedge = Remove factor exposure:

$$r^{\text{hedged}} = r^e - \beta \times f = \alpha + \epsilon$$

Why hedge?

Lower vol → bigger position within risk budget

Example (MSFT, \$1M vol budget):

Unhedged position: ~\$94M

Hedged position: ~\$136M (1.4× larger!)

Appraisal Ratio

$$AR = \alpha / \sigma(\epsilon)$$

= Sharpe ratio of hedged portfolio

Connection to overall Sharpe:

$$SR^2_{\text{total}} = SR^2_{\text{market}} + AR^2$$

Alpha and market are orthogonal bets.

Portable alpha = sell β , keep α .

Pod shops require factor-neutral positions across multiple factors (zero net β).

Chapter 7

Portfolio Mathematics

Big idea: Portfolio variance is NOT the weighted average of variances. Covariances dominate. Diversification is the only free lunch.

Portfolio Algebra

Portfolio return: $R_p = W'R = \sum w_j r_j$

Expected return: $E[R_p] = W'E[R]$

Portfolio variance: $\text{Var}(R_p) = W'\Sigma W = \sum_i \sum_j w_i w_j \text{Cov}(r_i, r_j)$

Diversification in Action

50-asset portfolio:

50 variance terms vs. 2,450 covariance terms

Adding a MORE volatile asset can REDUCE portfolio vol if correlation is low enough.

US + Intl (70/30): lower vol than 100% US

Weight Constraints

$\sum w_j = 1$ (fully invested)

$w_j \geq 0$ (long-only constraint)

With risk-free asset:

$\sum w_j + w_{rf} = 1$

Short = negative w , Leverage = $\sum |w_j| > 1$

Chapter 8

Capital Allocation I

Big idea: Two-Fund Separation — all investors hold the same risky portfolio (MVE); they differ only in how much risk-free asset to mix in.

Mean-Variance Efficient Portfolio

$$\mathbf{W}^* = (\mathbf{1}/\gamma) \Sigma^{-1} \mathbf{E}[\mathbf{R}^e]$$

MVE weights: inverse covariance $\times$ expected excess returns

Two-Fund Separation

Optimal portfolio vol:

$$\mathbf{w}^* \times \sigma(r_p) = (\mathbf{1}/\gamma) \times \text{SR}$$

Conservative investor: small w^* , more cash

Aggressive investor: large w^* , same risky mix

"All you need is Sharpe" — the MVE portfolio maximizes SR, everyone just scales it.

Empirical Example (FF6)

Factor Sharpe Ratios (annual):

Mkt: 0.46 SMB: 0.17 HML: 0.33

RMW: 0.41 CMA: 0.40 MOM: 0.49

MVE portfolio (in-sample):

SR $\approx$ 1.17

⚠ But: massive overfitting risk (Ch 12)!

Chapter 9

Factor Model Estimation

Big idea: Estimation risk is the silent killer. Small perturbations in inputs cause wild swings in MVE weights. Regularization is essential.

Estimation Risk & Hedged Portfolios

The Problem: MVE weights = $\Sigma^{-1}E[R^e]$. Small errors in $E[R]$ or $\Sigma \rightarrow$ extreme, unstable weights.

The Fix: Shrink toward simpler models. Use factor structure to reduce the number of parameters.

Factor model covariance: $\Sigma = \beta \text{Var}(F) \beta' + \text{Var}(\epsilon)$ (far fewer parameters than full Σ)

Variance, Betas, Factor premia

- Past data is highly informative
- Lots of statistical judgements regarding estimation windows needs to be made

Alphas

- Economics implies it is transitory
- Why? It gets competed away
- Unless based on non-public information

Chapter 10

Cross-Sectional Equity Strategies

*Big idea: Sort stocks on a characteristic → long the top, short the bottom.
This is how every major factor (value, momentum, quality) is constructed.*

The Portfolio Construction Recipe



Signal–Diversification Tradeoff

Fewer groups (5): weaker signal, more diversified, lower vol.

Many groups (100): extreme signal, but concentrated, higher vol.

Sweet spot: 10–20 groups typically optimal.

Market-cap weighting preferred: auto-rebalances, minimizes turnover, easy to trade.

The Factor Zoo — Key Factors

Value

SR: 0.33

Book / Market

Cheap stocks outperform
(contrarian / distress risk)

Momentum

SR: 0.49

Past 12m return
(skip last month)

Winners keep winning
(6–12 month horizon)

Profitability

SR: 0.41

Revenue – COGS
/ Assets

Profitable firms earn more
(quality signal)

Investment

SR: 0.40

Asset growth

Low investment firms
outperform (empire building)

Size

SR: 0.17

Market cap

Small caps earn premium
(weakest factor)

Momentum: The Details

Mom signal: $\text{Mom}_{i,t-2} = P_{t-2} / P_{t-14} - 1$ (12-month cumulative, skip most recent month)

Return Horizons

Short-term (<1m): Reversal
Intermediate (3–12m): Momentum
Long-term (3–5yr): Reversal

Momentum is the intermediate effect.

Skip Month

Jegadeesh (1990): short-term reversal contaminates signal. Skipping last month removes microstructure noise.

Crashes

Daniel & Moskowitz (2015): Momentum crashes hard in high-vol periods (2008, 2020). Highly profitable but risky.

Chapter 11

Capital Allocation II

*Big idea: When MVE is unreliable, use bet-sizing heuristics.
Simpler models often beat optimal ones out-of-sample.*

Bet-Sizing Heuristics

Mean-Variance

$$w_i \propto \alpha_i / \sigma^2(\epsilon_i)$$

Optimal in theory; fragile in practice

1/N

$$w_i = (1/N) \times \text{sign}(\alpha_i)$$

Equal weight; ignores signal strength

Proportional

$$w_i \propto \alpha_i$$

Weight by alpha; ignores risk

Risk Parity

$$w_i \propto 1/\sigma(\epsilon_i)$$

Equal vol contribution; robust

Min Variance

$$w_i \propto 1/\sigma^2(\epsilon_i)$$

Assumes equal alphas; uses risk only

Shrinkage

$$w_i \propto \alpha_i / [(1-\tau)\sigma^2 + \tau\bar{\sigma}^2]$$

Blend MV toward simpler model

Rule of thumb: small universe $\rightarrow$ 1/N or risk parity. Large universe $\rightarrow$ shrinkage or MV with regularization.

Chapter 12

Performance Evaluation

Big idea: Overfitting is the #1 enemy. If you didn't hold out a test sample, your backtest is a fairy tale. Multiple testing makes it worse.

Alpha Testing & Overfitting

Alpha Test

$$r_t^e = \alpha + \beta r_{mkt}^e + \varepsilon_t$$

$$\text{Appraisal Ratio} = \alpha / \sigma(\varepsilon) = t_\alpha / \sqrt{T}$$

Minimum AR to be p-confident true AR > target:

$$\text{AR}_{\text{thresh}} = \text{AR}_{\text{target}} + z_p / \sqrt{T/12}$$

Overfitting in Action (MVE)

In-sample (1963–2012):

$$\text{SR} = 1.34, \alpha = 12.3\%/yr, t = 9.07$$

Out-of-sample (2013–2025):

$$\text{SR} = 0.54, \alpha = 3.2\%/yr, t = 0.95$$

SR degrades by >50% out of sample.

Diagnostics Toolkit

1. Sharpe Ratio & t-stat 2. Alpha & Appraisal Ratio 3. Cumulative returns & drawdown 4. Tail behavior ($\pm 3\sigma$ frequency) 5. Fraction to Half (robustness) 6. OOS test efficiency 7. Multiple testing correction

Multiple testing: 100 noise strategies, $t > 1.64 \rightarrow$ expect ~4 false positives. With 1 real signal (SR=1), 48-month test, $t > 2$:

Hit rate = 15.9% | Detection rate = 48.5%

Publication bias: HML SR ≈ 0.6 in-sample $\rightarrow \approx 0$ post-publication.

Sample Splitting: The Golden Rules

- 1 Never estimate and evaluate on the same sample.
- 2 Never use future data (lag everything).
- 3 Keep a hold-out sample. Don't peek until done tuning.
- 4 Document all discarded ideas (affects multiple testing).

Three-Way Split: Estimation (fit model) → Tuning (select hyperparams) → Hold-out (final test)

Chapter 13

Machine Learning in Finance

Big idea: Signal-to-noise is destiny. In low-SNR domains like finance, regularized simple models often beat complex ones.

The Central Challenge

$$\mathbf{R}_{i,t+1} = \mathbf{F}(\mathbf{X}_{i,t}) + \boldsymbol{\varepsilon}_{i,t} \quad \text{Signal} \approx 0.1\text{--}0.5\%/ \text{month} \bullet \text{Noise} \approx 10\% \bullet \text{SNR} \approx 0.01$$

Model Comparison (29 characteristics, ~200K stock-months)

Lasso (L1)	SR: 1.03	Baseline — hard to beat!
Elastic Net	SR: 1.10	Handles correlated features
Lasso + interactions	SR: 2.40	435 features, marginal gain
Random Forest	SR: 1.47	Auto interactions, shallow
Gradient Boosted Trees	SR: 1.82	Sequential residual fitting
Neural Net (regularized)	SR: 2.61	Dropout + early stopping

ML Lessons for Finance

Start simple

Lasso/Elastic Net are hard to beat, interpretable, and robust. Complexity rarely pays in low-SNR.

Regularize everything

Dropout + weight decay + early stopping for NNs. Keep trees shallow (depth ≤ 3). Shrink covariances.

Respect the time series

Never use future data. Three-way split. Learning curves matter more than architecture.

Double descent

Overparameterized models can recover signal if it exists — but CANNOT create signal from noise.

Sample Questions from the Final

- After having the exam ready for Monday, I selected randomly 4 questions to share
- I will of course now have to prepare new questions!
- But this should also give a good estimate of what to expect
- By construction they are sampled from the same data generating process 😊
- Lets go through it

1) Multiple choice — explain your answer (30 pts)

For each question, **circle as many answers as needed** *and* **provide a one- or two-sentence explanation**. The explanation is required for full credit.

1.1) Which allocation rule is robust to noise in your estimates of expected returns?

MVE / tangency portfolio

Equal-weight

Inverse-volatility (diagonal risk parity)

Minimum-variance portfolio

1.2) A fund's CAPM alpha is 4%. After adding HML to the regression, alpha drops to 1%. What can you conclude?

The fund manager has 4% of true skill.

About 3% of the apparent alpha was actually a value tilt.

The fund has zero true skill.

Adding factors should not change the alpha.

1.3) You test 50 candidate signals at the 5% significance level. Under the null that none has alpha, the expected number of signals flagged as significant is approximately:

0

2.5

5

50

A pod asks you for \$50 million. They show a four-year track record: annual return 10%, vol 10%, SR 1. The market SR is 0.5 with a vol of 15%

Suppose you leverage this strategy to have vol equal to the market, what expected return you expect to get?

Your CIO says: "This looks twice as good as the market." Is he right? **Explain**

Name one non-statistical question about the strategy you would ask before allocating capital

In Assignment 5 you found that perturbing one industry's expected excess return by ± 1.96 standard errors (still within its 95% CI) caused the **tangency weights** to change dramatically — but the **in-sample Sharpe ratio** barely moved.

Explain in one or two sentences why the weights move so much.

Explain in one or two sentences why the in-sample SR barely moves under these perturbations.

You sort six stocks on a quality signal. Higher signal means higher quality. You form an equal-weighted long-short portfolio: long the top two stocks and short the bottom two stocks.

- a. Which stocks are in the long leg and which are in the short leg?
- b. Compute the long-leg return.
- c. Compute the short-leg return.
- d. Compute the long-short return.
- e. Compute the difference between the average signal in the long leg and the average signal in the short leg. Why is this useful?

Stock	Quality signal	Return next month
A	0.10	-3%
B	0.20	1%
C	0.40	0%
D	0.60	2%
E	0.80	4%
F	0.90	-1%